Software Documentation

1. Software Overview

Platforms used:

- Micropython: used for the input, output and processing of our system
- MIT App Inventor: used for the app interface, to ‘alert’ the micropython code to perform specific functions

2. System Logic

a) **Gateway one:**

1. User presses ‘OTP Generation Button’
2. App sends a variable via bluetooth
3. ESP32 receives variable; if condition is fulfilled
4. Random OTP is generated and displayed on app screen
5. When push button is pressed value is sent to the ESP32; compared with the OTP digits;
6. If condition fulfilled, servo motor moves; neopixel lights green, buzzer sounds
7. If condition not fulfilled, servo doesnt move; neopixel lights red, buzzer sounds

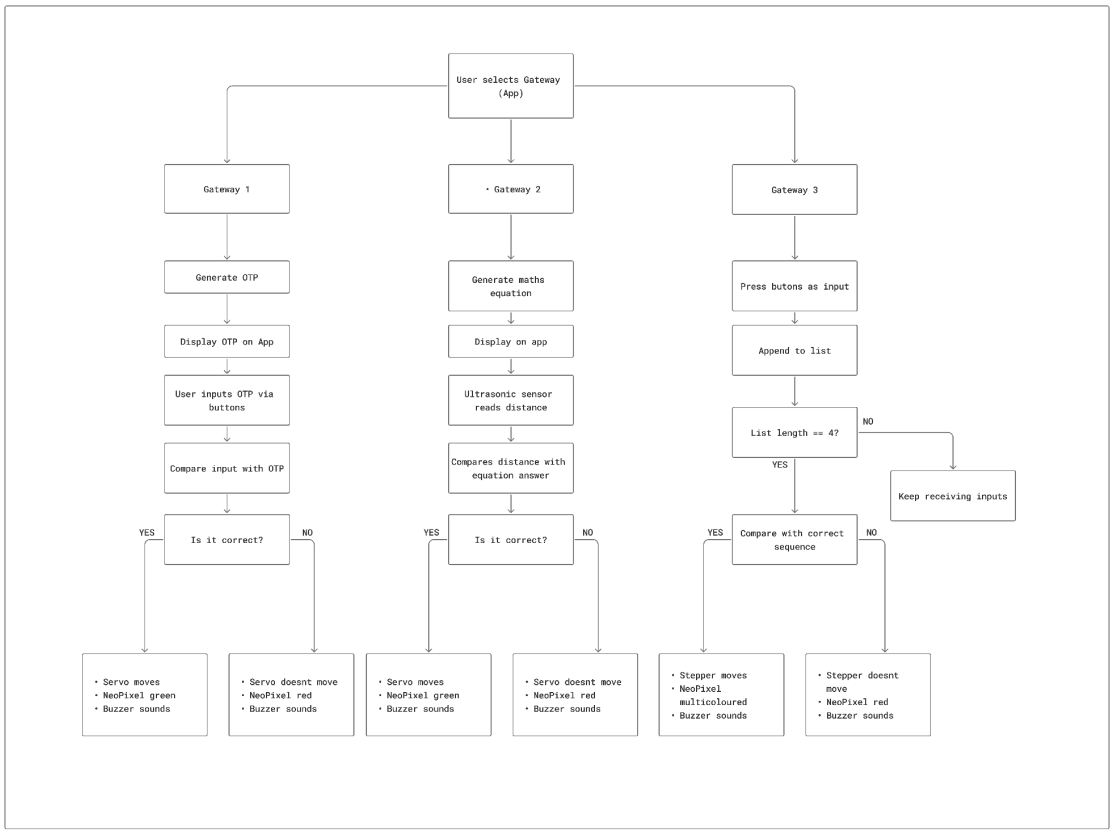
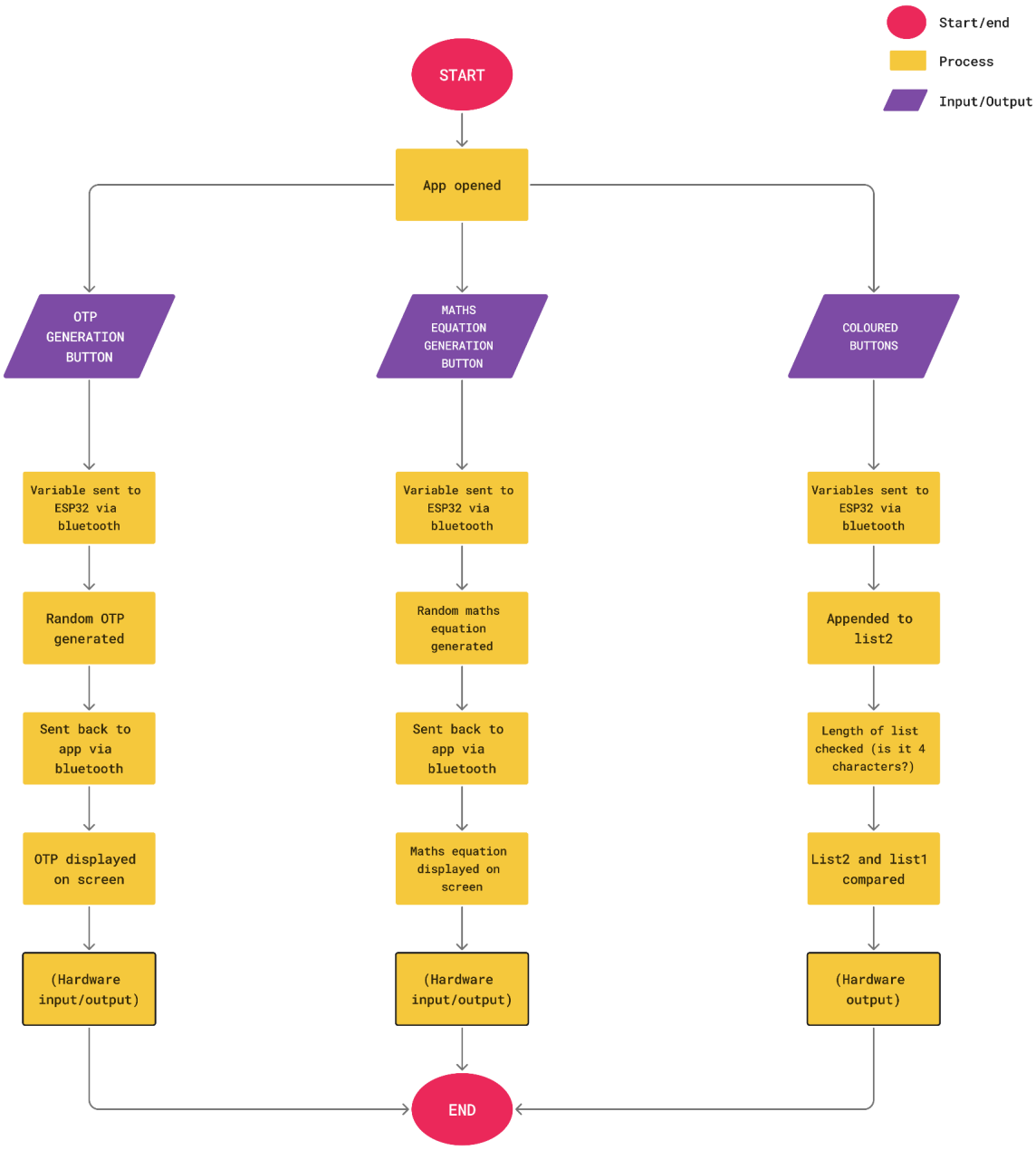
b) **Gateway two:**

1. User presses ‘Math Equation Generation Button’
2. App sends a variable via bluetooth
3. ESP32 receives variable; if condition is fulfilled
4. Random math equation is generated and displayed on app screen
5. Hand is placed some distance from ultrasound sensor; sends value to the ESP32
6. If condition fulfilled, servo motor moves; neopixel lights green, buzzer sounds
7. If condition not fulfilled, servo doesnt move; neopixel lights red, buzzer sounds

c) **Gateway three:**

1. User presses coloured buttons
2. App sends a variable with each button pressed
3. ESP32 receives variable, appends each to a list
4. If number of elements in the list is equal to 4, list is compared to another list having the actual sequence; if condition is fulfilled
5. If condition fulfilled, stepper motor moves; neopixel lights multicoloured, buzzer sounds
6. If condition not fulfilled, stepper does not move; neopixel lights red, buzzer sounds

3. Flowchart



4. App Flow (If MIT App Inventor)